Edge-Conditioned Feature Transform Network for Hyperspectral and Multispectral Image Fusion

Yuxuan Zheng, *Student Member, IEEE*, Jiaojiao Li, *Member, IEEE*, Yunsong Li, *Member, IEEE*, Jie Guo, Xianyun Wu, Yanzi Shi, *Student Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Despite recent advances achieved by deep learning techniques in the fusion of low-spatial-resolution hyperspectral image (LR-HSI) and high-spatial-resolution multispectral image (HR-MSI), it remains a challenge to reconstruct the high-spatial-resolution HSI (HR-HSI) with more accurate spatial details and less spectral distortions, since the low-level structure information such as sharp edges tends to be weakened or lost as the network depth grows. To tackle this issue, we creatively propose an edge-conditioned feature transform network (EC-FTN) in this article, which is mainly composed of three parts, namely, feature extraction network (FEN), feature fusion and transformation network (FFTN), and image reconstruction network (IRN). First, two computationally efficient FENs with 3-D convolutions and reshaping layers are employed to extract the joint spectral-spatial features of input images. Then, the FFTN conditioned on the edge map prior can fuse and transform the features adaptively, in which a fusion node and several cascaded feature modulation modules (FMMs) equipped with feature-wise modulation layers are constructed. Specifically, the edge map is generated via transfer learning, i.e., by applying the Sobel operator to feature maps of the red-green-blue (RGB) version of HR-MSI resulting from the pretrained VGG16 model without extra training. Finally, the desired HR-HSI is recovered from the transformed features through IRN. Furthermore, we elaborately design a weighted combinatorial loss function consisting of mean absolute error, image gradient difference, and spectral angle terms to guide the training. Experiments on both ground-based and remotely sensed datasets demonstrate that our EC-FTN outperforms state-of-the-art methods in visual and quantitive evaluations, as well as in fine details reconstruction.**

***Index Terms*— Edge prior, feature transform network, hyperspectral image (HSI), image fusion, multispectral image (MSI), transfer learning (TL).**

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Yuxuan Zheng, Yunsong Li, Jie Guo, Xianyun Wu, and Yanzi Shi are with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: yxzheng24@163.com; ysli@mail.xidian.edu.cn; jguo@mail.xidian.edu.cn; xywu@mail.xidian.edu.cn; yzshi_xidian@163.com).

Jiaojiao Li is with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China, and also with the CAS Key Laboratory of Spectral Imaging Technology, Xi'an 710119, China (e-mail: jjli@xidian.edu.cn).

Jocelyn Chanussot is with LJK, Inria, CNRS, Grenoble INP, Universite Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

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I. Introduction

HYPERSPECTRAL imaging which acquires images carrying rich spectra across a wide range of wavelengths has been a flourishing technology in the remote sensing (RS) domain. The obtained hyperspectral image (HSI) with high spectral resolution contributes greatly to a variety of applications, such as scene classification [1], target detection [2], and spectral unmixing [3]. Unfortunately, there is a critical tradeoff between the spectral and spatial resolutions provided by the existing hyperspectral (HS) imaging system due to various hardware restrictions. To guarantee a sufficient signal-to-noise-ratio (SNR), photons have to be collected in a much larger region on the imaging sensor, thus resulting in a low spatial resolution of the observed HSI [4]. This sacrifice of spatial resolution limits the interpretation and analysis of HSI scenes to some extent [5]. On the contrary, the multispectral image (MSI) can be captured by multispectral (MS) sensors in a significantly higher spatial resolution and a relatively lower spectral resolution. Therefore, fusing a low-spatial-resolution HSI (LR-HSI) with the corresponding high-spatial-resolution MSI (HR-MSI), termed as HS-MS fusion, has become a very popular scheme to enhance the spatial resolution of HSI, which has shown improved performance in many applications [6].

With the increasing availability of higher spatial resolution auxiliary images, a great number of fusion-based approaches have been developed to generate the high-resolution HSI (HR-HSI) in the past decades, which could be roughly divided into two classes. One category named HS pansharpening [7] is to fuse the LR-HSI with a high-resolution panchromatic (HR-PAN) image of the same scene. However, due to the very limited spectral resolution of a single-band PAN image, HS pansharpening techniques usually result in considerable spectral distortions [8]. Another called HS-MS fusion refers to the fusion of HS and MS images, which has gained tremendous attention recently. Unlike HS pansharpening, HS-MS fusion approaches can generally yield higher spectral quality owing to the spectral information of HR-MSIs [9]. Among the available conventional methods

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for HS-MS fusion, representative techniques can be broadly categorized as follows: component substitution (CS) [10], multiresolution analysis (MRA) [11], [12], and subspace-based approaches [13]–[19]. A comparative review of these fusion techniques can be found in [20], where the aforementioned ten methods [10]–[19] are compared through extensive experiments.

In recent years, motivated by the powerful ability of the deep learning (DL) technology in extracting hierarchical features and performing image super-resolution (SR), the DL techniques such as convolutional neural network (CNN) and residual network (ResNet) have been introduced to solve the problems of both HS pansharpening [21]–[24] and HS-MS fusion [25]–[32], which have attained rapid progress in fusion performance. However, the important low-level structure information like edges and corners is easy to be weakened or lost when networks go deeper [33], thus the reconstruction of HR-HSI with accurate high-frequency details and high spectral fidelity remains to be a challenging issue. Typically, the current DL-based HS image fusion methods suffer from several drawbacks as follows.

1) Many existing DL-based methods just treat the fusion process as a regression problem and simply adopt the flourished SR frameworks to learn a mapping from the directly stacked source images to the target HR data, which do not fully consider and exploit the prior knowledge of input, especially the low-level edge detail information of HR images that may be lost during fusion.
2) Both HS and MS images are volumetric data delivering 2-D spatial and 1-D spectral information simultaneously. Therefore, it is of crucial significance to adequately utilize the spatial-spectral structural property for these 3-D data with low cost. Nevertheless, most previous DL-based methods perform the fusion through using either just 3-D-CNN [25] that can lead to high computational complexity or two-branches 1-D-2-D CNN [26] which may fail to explore more spatial information when extracting spectral features from the HSI [34]. Moreover, the intrinsic structure information of input data may be lost during feature extraction.
3) Generally, the majority of fusion networks are optimized by minimizing the Euclidean distance-based loss function like mean square error (MSE), which tends to neglect the HS image fusion task-specific goals of spatial details enhancement and spectral characteristics preservation, resulting in relatively blurry predictions and certain spectral distortions.

To tackle the above issues, we propose a novel edge-conditioned feature transform network (EC-FTN) for HS-MS fusion in this article. By adaptively incorporating the edge prior information of HR-MSI into a feature-level fusion framework, our EC-FTN is capable of reconstructing the HR-HSI with fine details and high accuracy, as demonstrated in Fig. 1. To the best of our knowledge, this is the first attempt to investigate the HS-MS fusion problem through feature-wise modulation conditioned on the edge prior. In summary, the contributions of this work are mainly threefold:

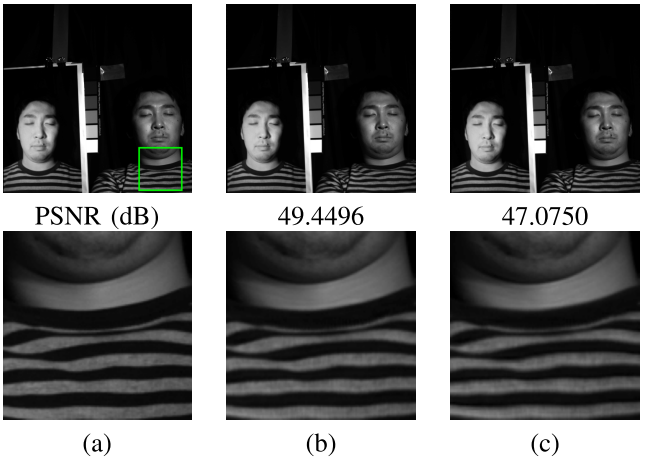


Fig. 1. Example of fusion results for *photo_and_face* (an HSI from the CAVE[1] dataset) demonstrating the effectiveness of our EC-FTN. The top and bottom rows show the 22nd band of reconstructed images and the demarcated areas zoomed in 4 times for easy comparison, respectively. The corresponding PSNR value (in dB) is shown under each reconstructed image. (a) Ground truth. (b) Proposed EC-FTN. (c) Proposed method without edge prior. Our proposed EC-FTN conditioned on the edge prior can achieve higher accuracy of spatial details reconstruction.

1) A novel EC-FTN with edge-conditioned feature-wise modulation layers (EC-FMLs) is proposed to perform HS-MS fusion at the feature level, where the edge map of HR-MSI generated by powerful transfer learning (TL) is served as an effective guidance for boosting the fusion accuracy. More specifically, to generate the robust edge map prior, we apply the Sobel operator to feature maps of the red-green-blue (RGB) version of HR-MSI resulting from the pretrained VGG16 [35] model without extra training.
2) Unlike previous DL-based HS-MS fusion architectures using only 3-D-CNN or 1-D-2-D CNN, we propose to adopt a lightweight hybrid 3-D-2-D CNN in our EC-FTN to explore the spatial and spectral features of input HS and MS images simultaneously, where 3-D and 2-D convolutions share feature maps by the reshaping operation to exploit more spatial information. In this way, the hybrid 3-D-2-D CNN is able to not only facilitate the joint spectral-spatial feature extraction and fusion without information loss or structure destruction but also ease the training burden in an elegant and efficient fashion.
3) Instead of directly utilizing the standard MSE-oriented objective function to minimize the error in early studies, we elaborately design a new weighted combination of mean absolute error (MAE), image gradient difference (IGD), and spectral angle (SA) regularization terms to guide the training of EC-FTN, which greatly contributes to reconstructing accurate spatial details and restraining spectral distortions simultaneously. Experiments on both ground-based and remotely sensed HS datasets demonstrate the superiority of our EC-FTN with the proposed weighted combinatorial loss function, particularly in terms of preserving edge details.

[1] http://www.cs.columbia.edu/CAVE/databases/

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II. Related Work

In this section, we briefly review some representatives closely relevant to ours, including DL-based methods for HS pansharpening and HS-MS fusion, as well as conditioning methods for CNNs.

A. DL-Based HS Image Fusion Methods

In the past few years, the enhancement of RS image spatial resolution has been widely investigated via DL methodology. In an earlier attempt, Masi *et al.* [36] proposed a pansharpening CNN (PNN) for the traditional MS pansharpening problem. With the stacked upsampled MSI and PAN image as input, the PNN was composed of a three-layer CNN that was similar to super-resolution convolutional neural network (SRCNN) [37]. Following the success of PNN, the spectrally predictive CNN structure for HS pansharpening named HyperPNN was presented in [21], where the spectral prediction capability was strengthened. By utilizing the Laplacian pyramid super-resolution network (LapSRN) [38] to conduct SR of the LR-HSI, Li *et al.* [22] proposed the detail-based deep Laplacian pansharpening (DDLPS) method to improve the spatial resolution of HSI. In [23], the deep residual CNN (DRCNN) was designed to boost the performance of HS pansharpening. Different from the above introduced methods which treat the features of input images equally, very recently, Zheng *et al.* [24] proposed a dual-attention residual network (DARN) for HS pansharpening, which can selectively learn more informative spatial-spectral features simultaneously to improve the fusion accuracy. For HS-MS fusion, a 3-D-CNN method was presented in [25], where principal component analysis (PCA) was used for the dimensionality reduction of HSI prior to the fusion. While this step could simplify the computational processing, it may inevitably destroy the structure information of HS data and hinder the fusion performance. According to different characteristics of the two image types, Yang *et al.* [26] proposed a CNN with two branches for HS-MS fusion, where 1-D CNN and 2-D CNN were, respectively, utilized to extract spectral and spatial information of HSI and MSI. However, the important spatial information of HSI may not be well explored in the two-branches 1-D-2-D CNN model. By combining the image priors learned via CNN into the fusion framework, a deep HSI sharpening (DHSIS) method was proposed in [27]. Taking the observation models for MS and HS images into consideration, Xie *et al.* [28] constructed an interpretable MS/HS fusion network, which unfolded the designed iterative algorithm as network layers. In [29], an HSI and MSI multiscale fusion network (HAM-MFN) was proposed, where a RMSE, angle and Laplacian (RAP) loss (RAP loss) function made up of relative mean squared error (RMSE), angle, and Laplacian loss was presented. Additionally, unsupervised learning which does not require plenty of data for training is another important branch of DL-based HS-MS fusion approaches. Qu *et al.* [30] designed an unregistered and unsupervised mutual Dirichlet Net to solve the HS-MS fusion problem, which projected both modalities into the same statistical space to extract their correlated representations. By modeling a variational probabilistic autoencoder framework, the FusionNet [31] was able to fuse the spatial and spectral information of the LR-HSI and HR-MSI in an unsupervised manner. In [32], Zhang *et al.* presented an unsupervised adaptation learning framework for HS-MS fusion, where the HR-HSI was recovered in a coarse-to-fine fashion by employing a fusion module and an adaptation module. While these DL-based approaches have made certain advances, to some extent, they may induce blurred results without fine spatial details and high spectral fidelity, since the informative low-level features of HR RS images such as edges are not well considered and utilized.

B. Conditioning Methods for CNNs

Recent years have witnessed great breakthroughs in vision-and-language tasks by introducing a variety of conditioning methods to neural networks. As a generalization of conditional normalization (CN) that replaces the parameters of feature-wise affine transformation in batch normalization (BN) [39], the Feature-wise Linear Modulation (FiLM) [40] layer has proven highly effective in image stylization [41], visual question answering [42] and speech recognition [43], which applies an affine transformation, i.e., per-channel scaling and shifting, to the network's intermediate features $\mathbf{f}$ with conditioning information $\mathbf{c}$. The formulation of FiLM can be written as

$$\mathrm{FiLM}(\mathbf{f}) = \gamma(\mathbf{c}) \times \mathbf{f} + \beta(\mathbf{c}) \tag{1}$$

where $\gamma(\mathbf{c})$ and $\beta(\mathbf{c})$ with dimensions equal to the channel number of condition $\mathbf{c}$ are the scaling and shifting parameters, respectively. For CNNs, however, since each channel of $\mathbf{c}$ is mapped into one scaling vector and one shifting vector for each feature map of $\mathbf{f}$ in a single FiLM layer, FiLM cannot deal with conditions that possess spatial information.

Considering that spatial information is critical for low-level image processing tasks such as SR, Wang *et al.* [44] proposed the spatial feature transform-based generative adversarial network (SFT-GAN) for color image SR, where the SFT layers conditioned on segmentation probability maps were designed. With preserved spatial dimensions of the feature maps, the SFT layer was capable of carrying out spatial-wise transformation, which enabled the SR network to recover more realistic textures. Borrowing the main idea of SFT-GAN [44] to the RS community, Zhang *et al.* [45] proposed a GAN architecture for the task of MS pansharpening, which was equipped with SFT layers under constraints of spectral features. Owing to the integration of MS information into the GAN framework in [45], satisfactory pan-sharpened results could be obtained. In our EC-FTN approach, with the consideration that spatial edge and detail information is crucial for reconstructing the high-frequency components of the HR-HSI, a new FTN conditioned on the edge prior is proposed, which elaborately takes advantage of the informative edge guidance to facilitate the performance of HS-MS fusion.

III. Proposed EC-FTN

A. Problem Formulation

The goal of HS-MS fusion is to fuse a LR-HSI with an HR-MSI of the same scene to generate an HR-HSI that has both high spatial and high spectral resolution. Specifically,

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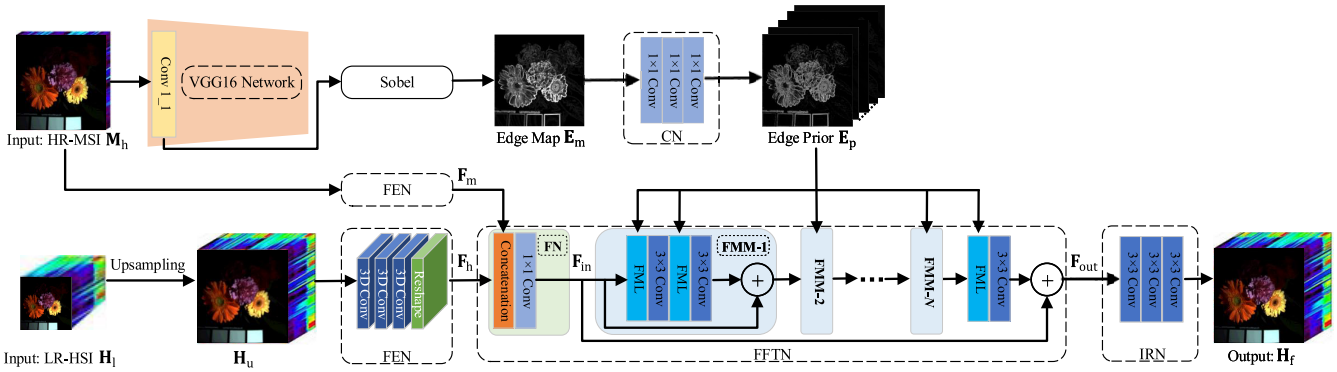


Fig. 2. Architecture of our proposed EC-FTN for the fusion of LR-HSI and HR-MSI.

let $\mathbf{H}_l \in \mathbb{R}^{x \times y \times Z}$ denote the acquired LR-HSI with a spatial dimension of $x \times y$ and Z spectral bands. $\mathbf{M_h} \in \mathbb{R}^{X \times Y \times z}$ $(X > x, Y > y, z < Z)$ denotes the available HR-MSI, where X and Y represent the sizes of two spatial modes and z is the band number. According to the Wald's protocol [46], the LR-HSI $\mathbf{H}_l$ and HR-MSI $\mathbf{M}_h$ can be, respectively, modeled as degraded versions of the ground truth HR-HSI $\mathbf{H}_g \in \mathbb{R}^{X \times Y \times Z}$ along the spectral and spatial dimension. Given $\mathbf{H}_l$ and $\mathbf{M}_h$, the fusion aims at estimating a fused HR-HSI $\mathbf{H}_f \in \mathbb{R}^{X \times Y \times Z}$, which is as similar as possible to the ground truth $\mathbf{H}_g$. In general, to obtain the estimated $\mathbf{H}_f$, many current DL-based methods leverage a network to learn the mapping function in a straightforward way

$$\mathbf{H}_f = \mathcal{F}_\theta(\uparrow \mathbf{H}_l, \mathbf{M}_h) \tag{2}$$

where $\uparrow$ stands for the upsampling operation, and $\mathcal{F}_\theta(\cdot)$ represents the fusion network parametrized by θ. Different from (2), by introducing the edge prior of HR-MSI to the fusion procedure, the generation of $\mathbf{H}_f$ in our EC-FTN can be reformulated as

$$\mathbf{H}_f = \mathcal{F}_\theta\big(\uparrow \mathbf{H}_l, \mathbf{M}_h \mid \mathbf{E}_p\big) \tag{3}$$

where $\mathbf{E}_p$ denotes the edge prior information that is beneficial for reconstructing the HR-HSI accurately.

B. Network Design

The overall architecture of our EC-FTN is shown in Fig. 2, which is mainly composed of three parts including two feature extraction networks (FENs), a feature fusion and transformation network (FFTN) and an image reconstruction network (IRN).

1) Feature Extraction Network: In the beginning, we first upsample the LR-HSI $\mathbf{H}_l$ to the same spatial scale as the HR-MSI $\mathbf{M}_h$ by utilizing a polynomial kernel with 23 coefficients [47], which is a classical interpolation method widely used in many RS image fusion works owing to its simplicity and fast implementation [48]. Then, we employ two FENs to extract features of the input MSI $\mathbf{M}_h$ and the upsampled HSI $\mathbf{H}_u$

$$\mathbf{F}_m = \mathcal{F}^1_{\text{FEN}}(\mathbf{M}_h), \quad \mathbf{F}_h = \mathcal{F}^2_{\text{FEN}}(\mathbf{H}_u) \tag{4}$$

where $\mathbf{F}_m$ and $\mathbf{F}_h$, respectively, denote the extracted features of $\mathbf{M}_h$ and $\mathbf{H}_u$. $\mathcal{F}^1_{\text{FEN}}$ and $\mathcal{F}^2_{\text{FEN}}$ represent the two FENs that have similar architectures. Taking $\mathcal{F}^2_{\text{FEN}}$ as an example, in each FEN, there are three serialized 3-D convolutional (Conv) layers with kernels of dimension $3 \times 3 \times 3$ to get the joint spectral-spatial feature representation of $\mathbf{H}_u$, followed by one reshaping layer. Although the 3-D Conv layer could extract discriminative features from both spectral and spatial dimensions, the inevitability of increased computational complexity and resource consumption in 3-D-CNN should also be considered, especially for the high-dimensional HS data. With these considerations in mind, we innovatively propose to adopt a lightweight hybrid 3-D-2-D CNN by introducing the reshaping layer at the end of FEN. Let $\mathbf{F}_{h-1} \in \mathbb{R}^{H \times W \times D \times C}$ denote the input tensor of the reshaping layer, which is composed of C feature maps with the dimension of $H \times W \times D$. The reshaping layer is used to reshape $\mathbf{F}_{h-1}$ to the output $\mathbf{F}_h \in \mathbb{R}^{H \times W \times (D \times C)}$ that includes $D \times C$ feature maps. In this way, both the spectral and spatial features of HSI and MSI can be obtained efficiently, which also facilitates the network training and subsequent processing.

2) Feature Fusion and Transformation Network: To fuse the extracted features $\mathbf{F}_m$ and $\mathbf{F}_h$ containing complementary information of the input MSI and HSI effectively, we first construct a fusion node (FN) consisting of one concatenation operation layer followed by a 1×1 Conv layer. In the FN, $\mathbf{F}_m$ and $\mathbf{F}_h$ are first concatenated without information loss, and then fed into the Conv layer to be fused adaptively. The process of FN can be formulated as

$$\mathbf{F}_{\text{in}} = \mathcal{F}_{\text{FN}}(\mathbf{F}_m, \mathbf{F}_h) = f^{1\times1}_{\text{Conv}}([\mathbf{F}_m, \mathbf{F}_h]) \tag{5}$$

where $[\cdot]$ and $f^{1\times1}_{\text{Conv}}(\cdot)$, respectively, represent the concatenation and convolution operation, and $\mathcal{F}_{\text{FN}}(\cdot)$ denotes the function of the FN. $\mathbf{F}_{\text{in}}$ is the fused feature serving as the input of the first feature modulation module (FMM-1) for the succeeding feature transformation.

As shown in Fig. 2, there are totally N FMMs with $2N + 1$ feature-wise modulation layers (FMLs) in the FFTN. For each FMM, it mainly consists of two FMLs and two 3×3 Conv layers. Here we take the first FML as an example for illustration, since all FMLs display a similar behavior. The structure of our FML is detailed in Fig. 3. Given input feature maps $\mathbf{F}_{\text{in}}$, the FML conditioned on the edge prior $\mathbf{E}_p$ applies an affine transformation to $\mathbf{F}_{\text{in}}$ by a pair of modulation

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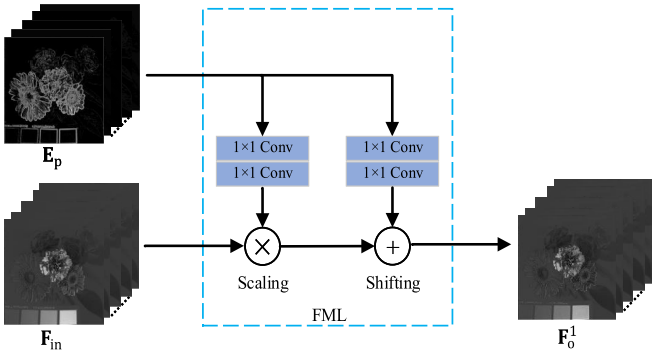


Fig. 3. Structure of the FML in our EC-FTN.

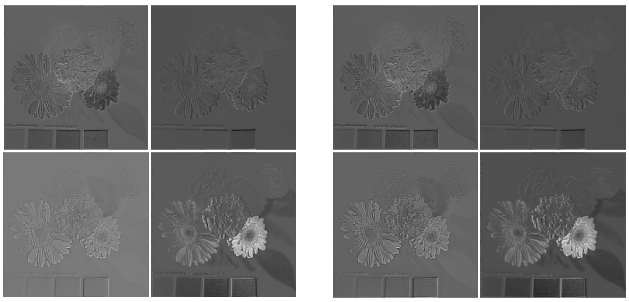


Fig. 4. Examples for illustrating the function of our EC-FML. We only show four representative feature maps for visualization. (a) Input feature maps of the EC-FML. (b) Output feature maps of the EC-FML.

parameters, i.e., the scaling and shifting parameters

$$\mathbf{F}_o^1 = \mathrm{FML}(\mathbf{F}_{\mathrm{in}}) = \alpha(\mathbf{E}_p) \otimes \mathbf{F}_{\mathrm{in}} + \eta(\mathbf{E}_p) \tag{6}$$

where $\otimes$ denotes the element-wise multiplication, and $\mathbf{F}_o^1$ is the output of the first FML. $\alpha(\mathbf{E}_p)$ and $\eta(\mathbf{E}_p)$ are scaling and shifting coefficients with the same dimension of $\mathbf{E}_p$ so that our FML is able to perform both feature-wise and spatial-wise transformation, which is different from FiLM [40]. Note that the leaky rectified linear unit (Leaky ReLU) activation layer with a parameter of 0.1 between the first 3×3 Conv layer and the second FML in each FMM is omitted in Fig. 2 for the sake of simplification. In addition, short and long skip connections are employed in each FMM and in the FFTN, respectively, which not only transmit the shallow features efficiently but also simplify the training process effectively. Based on the above description, the output of FFTN can be computed as

$$\mathbf{F}_{\mathrm{out}} = \mathbf{F}_{\mathrm{in}} + f_{\mathrm{Conv}}^{3\times 3}\left(\mathbf{F}_o^{2N+1}\right) \tag{7}$$

where $\mathbf{F}_o^{2N+1}$ denotes the modulated features of the $(2N+1)$th FML, and $f_{\mathrm{Conv}}^{3\times 3}(\cdot)$ represents the last Conv layer of the FFTN.

To better explore the function of our EC-FML, we visualize four representative feature maps from the input and output of the first EC-FML in Fig. 4. Compared with the input feature maps, the output features of EC-FML have much clearer structures in the edge regions and smaller pixel values of the smooth areas, which makes the high-frequency details more prominent, and thus further brings improvement in the fusion accuracy.

Next, we concentrate on the condition network (CN) which takes the edge map $\mathbf{E}_m$ as input and outputs the edge prior $\mathbf{E}_p$ shared by all FMLs. As displayed in Fig. 2, the CN mainly comprises three 1×1 Conv layers to provide intermediate conditions while alleviating the effects of interference in $\mathbf{E}_m$. To generate more robust $\mathbf{E}_m$, unlike traditional edge detection methods which employ operators directly on input images, we propose a simple yet powerful approach by application of the TL technique. Specifically, we apply the Sobel edge detector to feature maps of the three-channel RGB version of HR-MSI resulting from the pretrained VGG16 [35] network, which was previously trained on the ImageNet [49] dataset. Here, we choose VGG16 as the feature extractor for its simplicity and effectiveness. Since ImageNet is large enough for creating a generalized model, the well-trained VGG16 model has been demonstrated to be an effective feature extractor with a strong ability to generalize to new image datasets via TL [50], which is able to generate reliable and stable features for subsequent processing. Besides, based on the observation that low-level features such as edges are mainly extracted in shallow layers of CNN [51], we, therefore, consider utilizing the first Conv layer, i.e., Conv1_1 in VGG16, to get feature maps of the HR-RGB image without additional training, which are then fed into Sobel operator to obtain $\mathbf{E}_m$. As illustrated in Fig. 4, the generated edge map serves as reasonable and effective guidance for producing high-quality fusion results. The detailed procedure for the generation of $\mathbf{E}_m$ will be presented in Section IV-B1.

3) Image Reconstruction Network: After acquiring the fully fused and transformed features $\mathbf{F}_{\mathrm{out}}$, we exploit an IRN to recover the desired HR-HSI

$$\mathbf{H}_f = \mathcal{F}_{\mathrm{IRN}}(\mathbf{F}_{\mathrm{out}}) \tag{8}$$

where $\mathcal{F}_{\mathrm{IRN}}(\cdot)$ denotes the function of the IRN composed of three 3×3 Conv layers, as depicted in Fig. 2. $\mathbf{H}_f$ is the fused HR-HSI, which should be as similar as possible to the ground truth $\mathbf{H}_g$.

C. Loss Function

The loss function plays an indispensable role in supervised fusion networks. To address the issue that using only the conventional MSE loss may produce blurry results lacking high-frequency details [52], we design a new weighted combinatorial loss function $\mathcal{L}_{\mathrm{WC}}$ to train the proposed EC-FTN, consisting of three components as follows:

$$\mathcal{L}_{\mathrm{WC}} = \lambda_1 L_{\mathrm{MAE}} + \lambda_2 L_{\mathrm{IGD}} + \lambda_3 L_{\mathrm{SA}} \tag{9}$$

where L_{MAE}, L_{IGD}, and L_{SA} denote MAE loss, IGD loss and SA loss, respectively. λ_1, λ_2, and λ_3 are three weight coefficients corresponding to different components.

1) MAE Loss: Compared with MSE-based loss functions, the MAE loss has been verified to be a better choice in the HSI SR field [53]. Therefore, we adopt the MAE loss to minimize the pixel-wise error between the estimated HSI and the ground truth HR-HSI. For each fused image $\mathbf{H}_f$ and the corresponding ground truth $\mathbf{H}_g$, the MAE loss is defined as

$$L_{\mathrm{MAE}}(\mathbf{H}_g, \mathbf{H}_f) = \left\|\mathbf{H}_g - \mathbf{H}_f\right\|_1 \tag{10}$$

where $\|\cdot\|_1$ denotes ℓ_1-norm.

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2) IGD Loss: The image gradient can characterize the edge information of the image well. The smaller the IGD between $\mathbf{H}_g$ and $\mathbf{H}_f$, the higher the quality of edge details reconstruction. Therefore, to retain more high-frequency information and recover finer edge details, we develop the IGD loss as

$$L_{\mathrm{IGD}} = \left\|\nabla_1\mathbf{H}_g - \nabla_1\mathbf{H}_f\right\|_1 + \left\|\nabla_2\mathbf{H}_g - \nabla_2\mathbf{H}_f\right\|_1 \quad (11)$$

where $\nabla_1\mathbf{H}_g(h,w) = |\mathbf{H}_g(h+1,w) - \mathbf{H}_g(h,w)|$ and $\nabla_2\mathbf{H}_g(h,w) = |\mathbf{H}_g(h,w+1) - \mathbf{H}_g(h,w)|$ represent the gradients at pixel (h,w) of $\mathbf{H}_g$ along the horizontal and vertical directions, respectively. The definitions of $\nabla_1\mathbf{H}_f$ and $\nabla_2\mathbf{H}_f$ are similar to those of $\nabla_1\mathbf{H}_g$ and $\nabla_2\mathbf{H}_g$.

3) SA Loss: Apart from spatial consistency, spectral preservation is of significant importance for HS-MS fusion as well. This motivates us to construct the SA loss as

$$L_{\mathrm{SA}} = \frac{1}{\mathrm{XY}}\sum_{h,w}\arccos\left(\frac{\left\langle\mathbf{S}_{h,w},\hat{\mathbf{S}}_{h,w}\right\rangle}{\left\|\mathbf{S}_{h,w}\right\|_2\left\|\hat{\mathbf{S}}_{h,w}\right\|_2}\right) \quad (12)$$

where $\mathbf{S}_{h,w}$ and $\hat{\mathbf{S}}_{h,w}$ denote the spectral vector at the spatial location (h,w) of $\mathbf{H}_g$ and $\mathbf{H}_f$, respectively. $\langle\cdot,\cdot\rangle$ represents the inner product, and $\|\cdot\|_2$ is ℓ_2-norm.

By penalizing the pixel-level error, gradient-level difference, and SA dissimilarity between the ground truth and the predicted HSI simultaneously, the combined loss $\mathcal{L}_{\mathrm{WC}}$ encourages our EC-FTN to output the HR-HSI with accurate high-frequency details and reduced spectral distortions. The settings of three weight parameters and the effectiveness of our $\mathcal{L}_{\mathrm{WC}}$ will be discussed experimentally in Sections IV-D and IV-G4, respectively.

IV. Experiments and Discussions

A. Datasets and Metrics

To verify the effectiveness of our proposed EC-FTN for HS-MS fusion, one ground-based HS dataset named Columbia Computer Vision Laboratory (CAVE) [54] and two remotely sensed HS datasets, i.e., Pavia Center[2] (PC) and Washington dc Mall[3] (WDCM), were employed in the experiments. It is worthwhile pointing out that both ground-based and remotely sensed datasets with completely different spatial resolutions were utilized to test the performance of EC-FTN, which can conclusively demonstrate its robustness and applicability.

1) The CAVE dataset was obtained by a ground-based platform, consisting of 32 HSI scenes with a spatial size of 512 × 512 and 31 spectral bands covering the wavelength from 400 to 700 nm. The original images were used as the ground truth HR-HSIs to evaluate the fusion performance, and the LR-HSIs were simulated through Gaussian blurring and then downsampling the ground truth images via the factor of 4 × 4, that is, the spatial size of each LR-HSI in the CAVE dataset is 128 × 128. To create the three-band HR-MSI (RGB image), following [15], we used the Nikon D700 spectral response[4] to perform spectral degradation. In the training process, 22 HSIs from the CAVE dataset were randomly selected. The remaining ten images were served as the testing data.
2) The PC image originally containing 1096 × 1096 pixels was acquired by the airborne reflective optics system imaging spectrometer (ROSIS) sensor over the city of Pavia, Italy. After removing a 381-pixel-wide black band part and 13 noisy channels, the resultant remotely sensed data with 1096 × 715 pixels and 102 bands ranging from 430 to 860 nm has been widely used in the RS community. In our experimental design, as [23], [24] did, the PC dataset was comprised of 24 cubic image patches of size 160 × 160 × 102 without intersection, which were divided from the up-left 960 × 640 area of the scene. To synthesize the LR-HSI, we first filtered each cubic patch by a Gaussian blur and then downsampled it four times spatially. Thus the LR-HSI samples used for experiments had 40 × 40 spatial pixels with 102 spectral channels. The HR-MSI with four bands was simulated by utilizing the IKONOS-like reflectance spectral response filter [19]. In the PC dataset, 17 cubic patches were chosen randomly to form the training set, and the rest 7 patches were utilized for testing.
3) The WDCM image captured by the HS digital imagery collection experiment (HYDICE) sensor consisted of airborne HS data from the flight over the WDCM area. It contained 191 usable bands in the spectral region of 400–2400 nm, each of which had a spatial dimension of 1280 × 307 in the original data. For convenience, only the up-right 1200 × 240-pixel-size image was selected as the ground truth in our experiments, which was then partitioned into 20 nonoverlapped patches of size 120 × 120 × 191. With these patches serving as ground truth HSIs, the LR-HSIs were simulated through Gaussian blurring and spatial downsampling with a scale factor of 4. The six-band HR-MSIs were produced by using the spectral response of United States Geological Survey/National Aeronautics and Space Administration (USGS/NASA) Landsat 7 satellite [55]. We picked six patches at random from the WDCM dataset as the testing data, with the remainder used for training.

TABLE I
Characteristic and Statistic Information of Three Datasets

Dataset	Spatial Resolution	Spectral Resolution	Radiometric Resolution	Mean Values (Normalized)
CAVE	-	10 nm	16 bit	0.1194
PC	1.3 m	4 nm	14 bit	0.1475
WDCM	2.8 m	10 nm	16 bit	0.1544

In addition, some important characteristic and statistic information of the abovementioned three datasets are reported in Table I, including spatial, spectral, and radiometric resolutions, as well as mean values after data normalization. Note that in our experiments, the min–max normalization preprocessing was adopted to improve the training efficacy [56], i.e., the ground truth HR-HSIs were all scaled into the range of [0, 1].

[2]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Center_scene

[3]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

[4]https://www.maxmax.com/spectral_response.htm

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To assess the quality of fusion results quantitatively, five commonly used metrics are employed for quantitative evaluation, including peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [7], root mean squared error (RMSE), erreur relative globale adimensionnelle de synthèse (ERGAS) [20] and universal image quality index (UIQI) [57].

The PSNR reflects the spatial quality of each reconstructed band, and the average PSNR of all bands is usually used for the assessment of fusion results, that is

$$\mathrm{PSNR} = \frac{1}{Z}\sum_{b=1}^{Z}\mathrm{PSNR}\left(\mathbf{H}_g^b, \mathbf{H}_f^b\right) \tag{13}$$

where $\mathbf{H}_g^b$ and $\mathbf{H}_f^b$ denote the bth band image of the ground truth $\mathbf{H}_g \in \mathbb{R}^{X\times Y\times Z}$ and the fused image $\mathbf{H}_f \in \mathbb{R}^{X\times Y\times Z}$, respectively. A larger PSNR value indicates a better fusion quality.

The SAM is utilized to quantify the spectral distortion of each pixel (h, w), and the average SA value over all spatial pixels is computed for evaluation, that is

$$\mathrm{SAM} = \frac{1}{\mathrm{XY}}\sum_{h=1,w=1}^{\mathrm{XY}}\arccos\left(\frac{\left\langle \mathbf{S}_{h,w}, \hat{\mathbf{S}}_{h,w}\right\rangle}{\left\|\mathbf{S}_{h,w}\right\|_2\left\|\hat{\mathbf{S}}_{h,w}\right\|_2}\right) \tag{14}$$

where $\mathbf{S}_{h,w}$ and $\hat{\mathbf{S}}_{h,w}$ represent the spectral vector at pixel (h, w) of $\mathbf{H}_g$ and $\mathbf{H}_f$, respectively. $\langle\cdot,\cdot\rangle$ is the inner product, and $\|\cdot\|_2$ is ℓ_2-norm. The ideal value of SAM is 0.

The RMSE and ERGAS can provide global quality measures of the fusion results. The former is defined as

$$\mathrm{RMSE} = \frac{\|\mathbf{H}_g - \mathbf{H}_f\|_F}{\sqrt{\mathrm{XYZ}}} \tag{15}$$

where $\|\cdot\|_F$ is the Frobenius norm. Another global index ERGAS is expressed as follows:

$$\mathrm{ERGAS} = 100\frac{X}{x}\sqrt{\frac{1}{Z}\sum_{b=1}^{Z}\left[\frac{\mathrm{RMSE}_b}{\mu\left(\mathbf{H}_g^b\right)}\right]^2} \tag{16}$$

where RMSE_b and $\mu(\mathbf{H}_g^q)$ are the RMSE and mean of the bth band image, respectively. The optimal values of RMSE and ERGAS are both 0.

The UIQI is first calculated on a sliding window of size 32×32 and then averaged over the spatial domain at each band, which can be extended for HSI evaluation by averaging over all spectral bands. Let $\mathbf{D}_{b,a}$ and $\widehat{\mathbf{D}}_{b,a}$ represent the ath window of $\mathbf{H}_g^b$ and $\mathbf{H}_f^b$, the UIQI is defined as

$$\mathrm{UIQI} = \frac{1}{\mathrm{ZA}}\sum_{b=1,a=1}^{\mathrm{ZA}}\frac{2\sigma_{\mathbf{D}_{b,a}\widehat{\mathbf{D}}_{b,a}}}{\sigma_{\mathbf{D}_{b,a}}^2+\sigma_{\widehat{\mathbf{D}}_{b,a}}^2}\frac{2\mu_{\mathbf{D}_{b,a}}\mu_{\widehat{\mathbf{D}}_{b,a}}}{\mu_{\mathbf{D}_{b,a}}^2+\mu_{\widehat{\mathbf{D}}_{b,a}}^2} \tag{17}$$

where A is the number of windows, and $\sigma_{\mathbf{D}_{b,a}}$ and $\mu_{\mathbf{D}_{b,a}}$ are the standard deviation and mean of $\mathbf{D}_{b,a}$, respectively. $\sigma_{\mathbf{D}_{b,a}\widehat{\mathbf{D}}_{b,a}}$ is the sample covariance between $\mathbf{D}_{b,a}$ and $\widehat{\mathbf{D}}_{b,a}$. The UIQI varies within a range of $[-1, 1]$, with an optimal value of 1 when $\mathbf{H}_g$ and $\mathbf{H}_f$ are identical.

B. Implementation Details

1) Detailed Steps for Edge Map Generation: The generation of edge map $\mathbf{E}_m$ mainly involves three steps: 1) feature maps of the three-channel RGB version of input HR-MSI are computed by employing the Conv layers of VGG16 model with pretrained weights[5]; 2) Sobel operator is applied to the resultant features to get the gradient map of each filter in the Conv1_1 layer, which is the first Conv layer of VGG16; and 3) gradient maps of all filters in Conv1_1 are averaged to generate $\mathbf{E}_m$. Without any extra training and additional manual feature extraction, the robust edge map of HR-MSI can be obtained easily and efficiently.

2) Training and Testing Details: The training and testing of our proposed EC-FTN are performed using the TensorFlow framework on a GeForce GTX 1080Ti graphic processing unit (GPU). To prepare the training samples, we use a 32×32 sliding window to crop paired patches from the training sets. In the training stage, the zero-padding strategy is employed in all Conv layers to ensure that the spatial size of each feature map is kept unchanged. The batch size is set as 32 according to the limitation of GPU memory. We adopt the Adam algorithm with a learning rate of 1×10^{-4} to train our EC-FTN for 1.5×10^3 epochs. In the testing process, we feed each test pair $\{\mathbf{H}_u, \mathbf{M}_h \mid \mathbf{E}_p\}$ into the well-trained EC-FTN to get the fused HSI $\mathbf{H}_f$.

C. Determination of FMM Numbers in FFTN

The number of FMMs in the FFTN has an important effect on the fusion performance, which determines the depth of our EC-FTN. To study the impact of different FMM numbers, we conduct experiments on all the three datasets and list the average quantitative results of two global quality measures, i.e., RMSE and ERGAS, with different settings of N in Table II, where the best indexes are marked in bold. Apparently, for the three experimental datasets, most of fusion results obtain the best RMSE and ERGAS performance when $N = 5$. Therefore, the number of FMMs is set to $N = 5$ and thus there are 11 FMLs in total in the FFTN.

D. Selection of Weight Parameters in Loss Function

In the proposed loss function $\mathcal{L}_{\mathrm{WC}}$ of our EC-FTN method, three weight parameters λ_1, λ_2, and λ_3 used as tradeoffs need to be set. To make the MAE term play a dominant role, we first set $\lambda_1 = 1$. Moreover, it is found in preliminary experiments that L_{IGD} has a similar magnitude with L_{MAE}. Hence, λ_2 is also set as 1. In order to explore the effect of L_{SA} that determines the spectral constraint in our $\mathcal{L}_{\mathrm{WC}}$, we conduct experiments on three datasets to evaluate the influence of different λ_3 settings on the average quantitative performance of RMSE and ERGAS, which are two global quality measures. In our experiments, λ_3 is chosen from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. As shown in Table III, for the three experimental datasets, the RMSE results remain relatively stable with different values of λ_3. For the performance of ERGAS, it obtains the best result for the CAVE, PC, and WDCM dataset when λ_3 equals 0.001, 0.01, and 0.001, respectively. Therefore, we set $\lambda_3 = 10^{-3}$ for

[5]http://www.cs.toronto.edu/ frossard/post/vgg16/

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TABLE II
AVERAGE RMSE AND ERGAS PERFORMANCE WITH DIFFERENT SETTINGS OF N ON THREE DATASETS

Dataset	N	3	4	5	6
CAVE	RMSE (↓)	0.0034	0.0034	**0.0032**	0.0033
	ERGAS (↓)	0.8005	0.7977	**0.7577**	0.7772
PC	RMSE (↓)	0.0064	**0.0062**	**0.0062**	0.0063
	ERGAS (↓)	1.4765	1.4715	**1.4711**	1.4759
WDCM	RMSE (↓)	0.0026	0.0025	**0.0024**	0.0025
	ERGAS (↓)	1.8913	1.8287	**1.8136**	1.9412

TABLE III
AVERAGE RMSE AND ERGAS PERFORMANCE WITH DIFFERENT SETTINGS OF λ_3 ON THREE DATASETS

Dataset	λ_3	10^{-4}	10^{-3}	10^{-2}	10^{-1}
CAVE	RMSE (↓)	0.0034	**0.0032**	0.0036	0.0035
	ERGAS (↓)	0.8053	**0.7577**	0.8420	0.8412
PC	RMSE (↓)	0.0063	**0.0062**	**0.0062**	0.0064
	ERGAS (↓)	1.4832	1.4784	**1.4711**	1.5112
WDCM	RMSE (↓)	0.0027	**0.0024**	**0.0024**	0.0025
	ERGAS (↓)	1.8714	**1.8136**	1.8315	2.1767

the CAVE and WDCM dataset, and $\lambda_3 = 10^{-2}$ for the PC dataset.

E. Comparison Against State-of-the-Art Methods

We first evaluate our EC-FTN method on the above-mentioned three datasets in comparison with eight representative fusion approaches including: Gram–Schmidt adaptive (GSA) [10], generalized Laplacian pyramid (GLP) [12], coupled non-negative matrix factorization (CNMF) [13][6], International Conference on Computer Vision 15 (ICCV15) [16][7], HySure [14][8], 3-D-CNN [25], DHSIS [27][9], and DARN [24][10], representing the state-of-the-art traditional and DL-based methods. Note that GSA and GLP are originally proposed to solve the conventional MS pansharpening problem. Following [20], we apply GSA to HS-MS fusion by constructing multiple image sets, where each set is composed of one MS band and corresponding HS bands grouped by correlation-based clustering. To adapt the GLP scheme to HS-MS fusion effectively, we adopt the *hypersharpening* technique [58], which obtains a synthesized high-resolution image by linear regression of MS band images. For the implementations of GSA and GLP, we utilize the pansharpening toolbox[11] presented in [47], which can also contribute to fair comparisons and assessments of the competing fusion algorithms. The 3-D-CNN method was implemented by ourselves using Python language following the parameter settings in [25].

1) Performance Comparison on the CAVE Dataset: Table IV reports the average quantitative performance of competing methods on the ten testing images from the CAVE dataset, where the best results are highlighted in bold. As shown in Table IV, most DL-based methods perform better than the traditional methods in both spatial and spectral quality. Among DL-based methods, the 3-D-CNN obtains the worst evaluation values. This can be attributed to the comparatively shallow and simple network architecture of 3-D-CNN. And it is clearly seen that the proposed EC-FTN outperforms the other competing methods in terms of all the quantitative metrics.

TABLE IV
AVERAGE QUANTITATIVE PERFORMANCE OF COMPETING METHODS ON THE CAVE DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
GSA	40.7378	4.3812	0.0088	2.1945	0.9243
GLP	35.0571	3.3751	0.0163	3.7236	0.9320
CNMF	37.9286	4.5820	0.0130	3.0228	0.8832
ICCV15	40.6704	5.1663	0.0088	2.2289	0.9036
HySure	31.7767	6.4305	0.0230	5.7875	0.8166
3D-CNN	39.7193	3.5199	0.0105	2.2804	0.9138
DHSIS	47.8214	2.4960	0.0043	1.0120	0.9636
DARN	48.1963	2.3594	0.0039	0.9553	0.9691
Proposed	**50.3748**	**1.6547**	**0.0032**	**0.7577**	**0.9747**

Fig. 5 shows the fusion results and corresponding mean absolute difference maps of *stuffed_toys* (an HSI from the CAVE dataset) obtained by different methods. Each difference map reflects the error between a reconstructed HSI and the ground truth. The darker blue the map appears, the smaller the error is. In each error map, two regions where the differences are evident are marked in red rectangles for easy visual comparison. It can be observed from Fig. 5 that the GLP and HySure approaches produce obvious blur and color differences in the marked regions. In contrast, the proposed method can provide the clearest spatial details and the darkest error image, which means that our EC-FTN performs the best in visual comparison among these competing methods.

To illustrate the superior spectral preservation performance of our proposed method, we visualize the SAM results of different methods for *stuffed_toys* in Fig. 6. This visualization reflects the spatial distribution of spectral distortion. In each image, three meaningful areas are bounded by red rectangles for better comparison. Through comparing the marked regions, it can be obviously found that the proposed method achieves the smallest SAM among all the competing approaches, which indicates the superiority of our proposal in spectral fidelity preservation.

2) Performance Comparison on the PC Dataset: The average quantitative performance of different methods on the testing set from the PC dataset is listed in Table V. As can be seen from Table V, the proposed method outperforms others with regard to all the objective metrics, since it has the highest PSNR and UIQI values, and the lowest SAM, RMSE, and ERGAS values.

The fusion results of the 19th patch from the PC dataset named *PC_patch19* obtained by the competing methods are displayed in Fig. 7. For easy visual comparison,

[6] http://naotoyokoya.com/Download.html
[7] https://github.com/lanha/SupResPALM
[8] https://github.com/alfaiate/HySure
[9] https://github.com/renweidian/DHSIS
[10] https://github.com/yxzheng24/IEEE_TGRS_DHP-DARN
[11] http://openremotesensing.net/knowledgebase/a-critical-comparison-among-pansharpening-algorithms/

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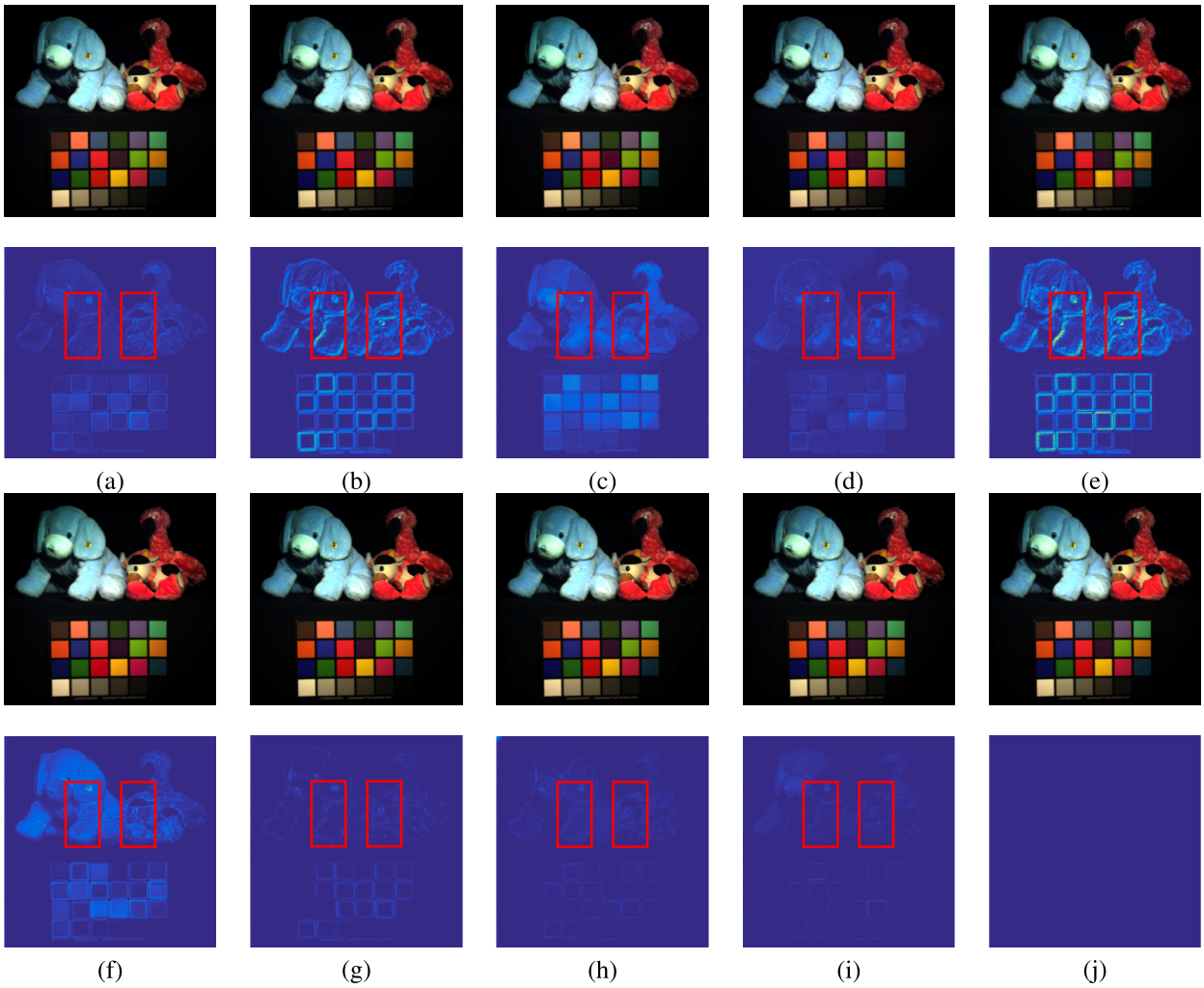


Fig. 5. First and third rows show the fusion results of *stuffed_toys* (an HSI from the CAVE dataset) obtained by different methods. We present the composite HSIs with bands 24-16-6 as R-G-B. The second and fourth rows show the corresponding mean absolute difference maps of different methods. (a) GSA. (b) GLP. (c) CNMF. (d) ICCV15. (e) HySure. (f) 3-D-CNN. (g) DHSIS. (h) DARN. (i) Proposed. (j) Ground truth.

TABLE V
AVERAGE QUANTITATIVE PERFORMANCE OF COMPETING METHODS ON THE PC DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
GSA	36.9405	4.8809	0.0162	2.7201	0.9697
GLP	32.9922	4.8170	0.0229	3.9056	0.9522
CNMF	35.5828	4.0197	0.0180	3.0717	0.9643
ICCV15	34.1676	4.5728	0.0191	3.6507	0.9534
HySure	36.9073	5.3369	0.0153	2.7926	0.9702
3D-CNN	36.6701	4.8093	0.0149	3.2311	0.9605
DHSIS	42.7467	3.4834	0.0088	1.8444	0.9831
DARN	45.6188	2.7566	0.0067	1.5600	0.9869
Proposed	**46.5093**	**2.5288**	**0.0062**	**1.4711**	**0.9888**

one demarcated area is zoomed in 4 times in each subimage. As depicted in Fig. 7, there are color distortions of the fusion results generated by most of the conventional approaches, such as GLP, CNMF, and ICCV15. Among the DL-based methods, blur details and spectral distortions can be easily found in the zoomed-in area of the image reconstructed by 3-D-CNN, and our proposed EC-FTN method can provide the fusion result with more accurate edges and less color differences than DHSIS and DARN, which is closest to the ground truth.

To compare the fusion performance in each spectral band, the PSNR curves of different methods for *PC_patch19* are shown in Fig. 8(a). Apparently, the proposed EC-FTN obtains larger PSNR values in most of the spectral bands compared with other methods, which verifies the higher fusion accuracy of our EC-FTN. In addition, Fig. 8(b) presents the curves of spectral reflectance differences between the ground truth and the fused images on pixel (40,44) of *PC_patch19* which is marked in red in Fig. 7(j), locating at the center of the demarcated area. A dark dotted line serving as the benchmark is drawn in Fig. 8(b). Being closer to the benchmark line means less spectral distortion. As shown in Fig. 8(b), the difference curve of the proposed EC-FTN method is closest to the dotted line, which demonstrates the outstanding spectral preservation property of our EC-FTN method.

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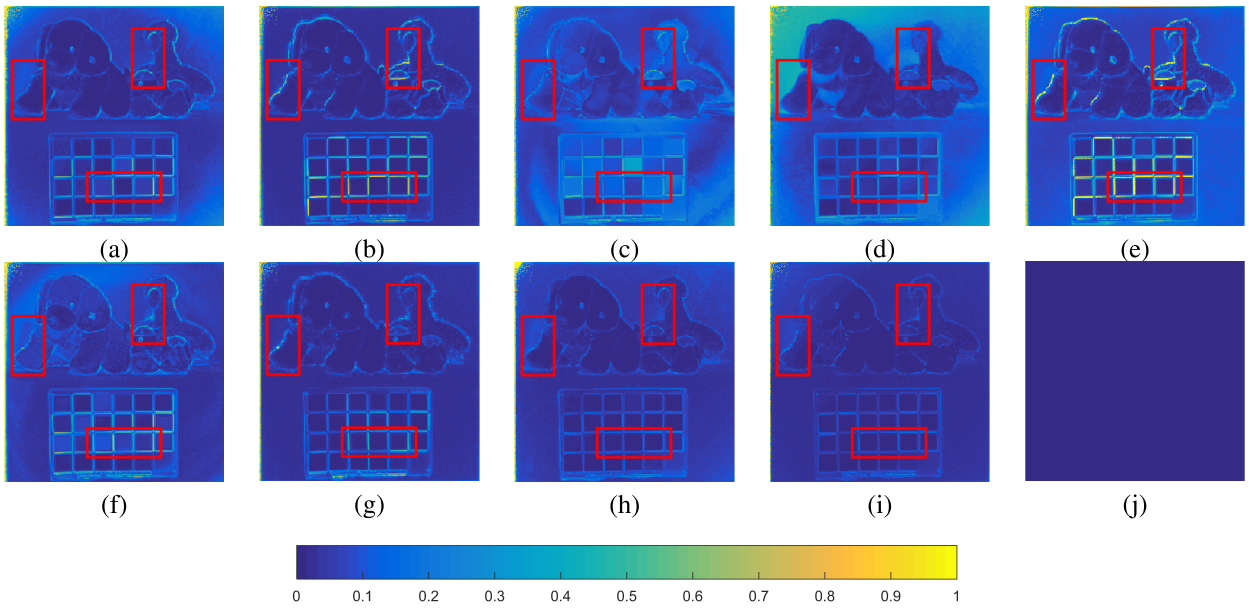


Fig. 6. Visualization of SAM results obtained by different methods for *stuffed_toys* (an HSI from the CAVE dataset). (a) GSA. (b) GLP. (c) CNMF. (d) ICCV15. (e) HySure. (f) 3-D-CNN. (g) DHSIS. (h) DARN. (i) Proposed. (j) Ground truth.

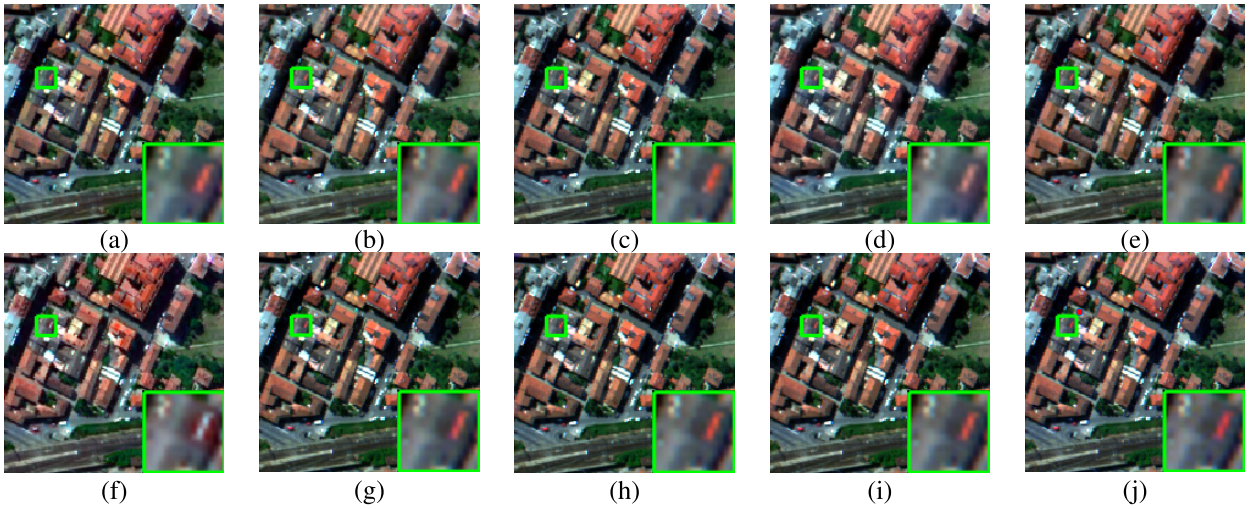


Fig. 7. Fusion results of *PC_patch19* (the 19th patch from the PC dataset) obtained by different methods, with demarcated areas zoomed in 4 times for easy comparison. We present the composite HSIs with bands 60-30-10 as R-G-B. (a) GSA. (b) GLP. (c) CNMF. (d) ICCV15. (e) HySure. (f) 3-D-CNN. (g) DHSIS. (h) DARN. (i) Proposed. (j) Ground truth.

3) Performance Comparison on the WDCM Dataset: The average PSNR, SAM, RMSE, ERGAS, and UIQI results of competing methods on the six testing data from the WDCM dataset are exhibited in Table VI. It can be obviously observed from Table VI that the proposed method surpasses all the other fusion algorithms in quantitative evaluations.

For visual comparison, Fig. 9 shows the fusion results of different methods with demarcated areas zoomed in 4 times for *WDCM_patch15*, which is the 15th patch from the WDCM dataset. As shown in Fig. 9, obvious spatial distortions can be seen in the enlarged regions of results obtained by GLP, CNMF, and HySure methods. Compared with the images reconstructed by the competing DL-based approaches, both the spatial sharpness and the color accuracy of the enlarged part in the fusion result generated by our proposal are most approximate to the ground truth.

To compare the fusion performance w.r.t. different spectral bands, the PSNR and spectral reflectance difference curves of different methods for *WDCM_patch15* are shown in Fig. 10. It can be seen from Fig. 10(a) that our EC-FTN performs better than the other approaches in almost all bands, especially in the last dozens of spectral channels. Fig. 10(b) displays the spectral difference values on pixel (8, 15) centered in the demarcated area of *WDCM_patch15*, which is marked in red in Fig. 9(j). According to Fig. 10(b), the proposal achieves a better approximation of the benchmark line, which indicates a more reliable performance of our EC-FTN method in the preservation of spectral information. The experimental results

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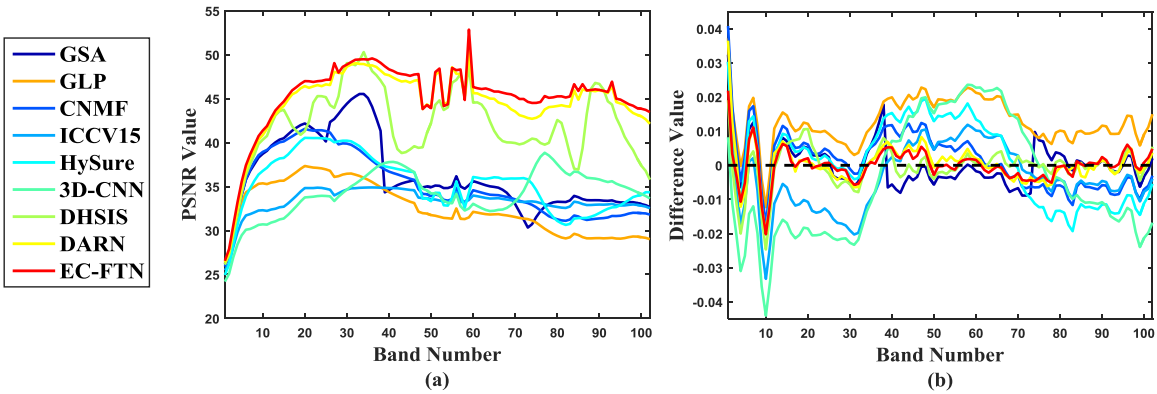


Fig. 8. PSNR and spectral reflectance difference curves of different methods for *PC_patch19*. (a) PSNR value. (b) Difference value on pixel (40, 44).

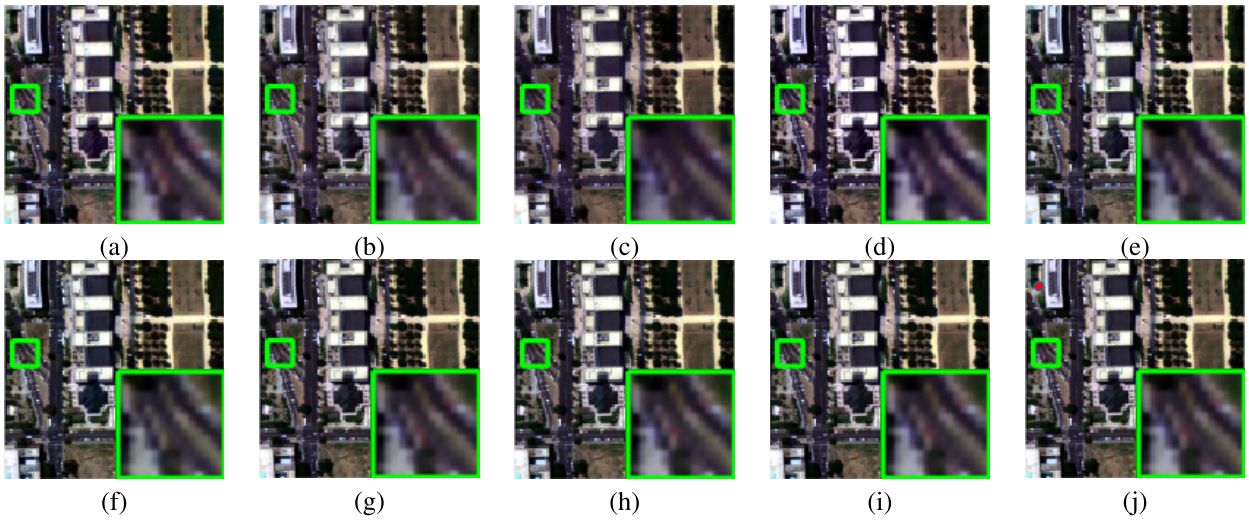


Fig. 9. Fusion results of *WDCM_patch15* (the 15th patch from the WDCM dataset) obtained by different methods, with demarcated areas zoomed in 4 times for easy comparison. We present the composite HSIs with bands 50-37-22 as R-G-B. (a) GSA. (b) GLP. (c) CNMF. (d) ICCV15. (e) HySure. (f) 3-D-CNN. (g) DHSIS. (h) DARN. (i) Proposed. (j) Ground truth.

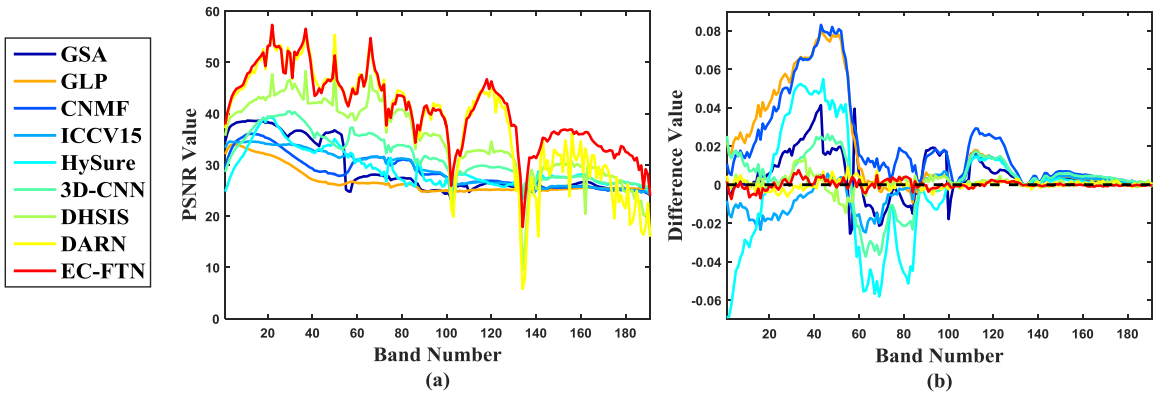


Fig. 10. PSNR and spectral reflectance difference curves of different methods for *WDCM_patch15*. (a) PSNR value. (b) Difference value on pixel (8, 15).

are shown in Figs. 9 and 10 fully showcase the superior fusion performance of our EC-FTN method.

F. Running Time

To investigate the computational burden of all the compared algorithms, we list the average running time of competing methods on the three experimental datasets in Table VII. Note that the experiments of the first five traditional methods are all conducted by using MATLAB with Intel Xeon E5-1620 3.50-GHz CPU, and the testing times of the last four DL-based methods are all measured by using TensorFlow with a GeForce GTX 1080Ti GPU. It can be observed from Table VII that

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TABLE VI

AVERAGE QUANTITATIVE PERFORMANCE OF COMPETING METHODS ON THE WDCM DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
GSA	31.3428	3.1702	0.0129	2.8384	0.9720
GLP	28.6689	3.3816	0.0186	3.3711	0.9649
CNMF	31.1700	2.2427	0.0126	2.7166	0.9724
ICCV15	30.1297	2.3223	0.0115	3.4034	0.9627
HySure	30.5039	3.5547	0.0125	3.1456	0.9571
3D-CNN	30.7223	2.5565	0.0103	3.3057	0.9626
DHSIS	39.1526	1.1364	0.0042	3.0422	0.9730
DARN	39.1447	0.7729	0.0026	3.0002	0.9765
Proposed	**41.4166**	**0.7163**	**0.0024**	**1.8136**	**0.9908**

TABLE VII

RUNNING TIME (S) OF COMPETING METHODS ON THREE DATASETS

Method	Dataset		
	CAVE	PC	WDCM
GSA	**0.4384**	**0.2070**	**0.2651**
GLP	5.9753	1.9356	2.2623
CNMF	17.5120	2.9595	2.1964
ICCV15	149.4450	6.1853	9.4468
HySure	312.2029	25.7862	13.6887
3D-CNN	**0.1070**	**0.0375**	**0.0351**
DHSIS	0.7109	0.2110	0.2168
DARN	0.2847	0.0493	0.0538
Proposed	0.4881	0.1107	0.1136

GSA costs less time than the other traditional methods for all three datasets. Among the DL-based methods, the running time of 3-D-CNN is the least. However, due to the shallow architecture of 3-D-CNN, the fusion performance of 3-D-CNN is usually relatively poor. On the contrary, the proposed EC-FTN method can obtain significant improvements in fusion accuracy with a slightly increased computational burden.

G. Ablation Study

Here, we study the contributions of elaborate designs in the proposed EC-FTN method through ablation experiments conducted on the PC dataset, including the lightweight hybrid 3-D-2-D CNN, the VGG16 network for edge map generation, the EC-FML, and the weighted combinatorial loss function.

1) Effect of the Lightweight Hybrid 3-D-2-D CNN: In the stage of feature extraction, we design a lightweight and generic framework by introducing the hybrid 3-D-2-D CNN with a reshaping layer in each FEN. To test the effect of the lightweight hybrid 3-D-2-D CNN, we compare it with the baseline design for feature extraction, termed as 2-D-CNN, which utilizes only 2-D Conv layers to extract features of the input MSI and the upsampled HSI, while keeping other settings the same. Another alternative is to use 3-D Conv layers only without any reshaping operation in each FEN. Note that this alternative method is not included in the comparison for reasons of high computation cost and limited GPU memory.

Table VIII lists the average quantitative performance of 2-D-CNN and hybrid CNN on the PC dataset. Clearly, compared with the baseline method, the hybrid CNN model achieves superior performance in terms of all the quantitative indicators, especially in reducing the spectral distortions. This is because the 3-D Conv layers in our FENs can extract the informative spectral-spatial features jointly, which benefits for a more accurate reconstruction of the desired HR-HSI. To demonstrate the effectiveness of the hybrid CNN visually, an example of fusion results for *PC_patch21* (the 21st patch from the PC dataset) is shown in Fig. 11. Again we see that our lightweight hybrid 3-D-2-D CNN does a better job at reconstructing spatial details and preserving spectral information.

TABLE VIII

AVERAGE QUANTITATIVE PERFORMANCE OF DIFFERENT FEN DESIGNS (2-D-CNN AND HYBRID CNN) ON THE PC DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
2D-CNN	46.3186	2.5606	0.0064	1.4952	0.9885
Hybrid CNN	**46.5093**	**2.5288**	**0.0062**	**1.4711**	**0.9888**

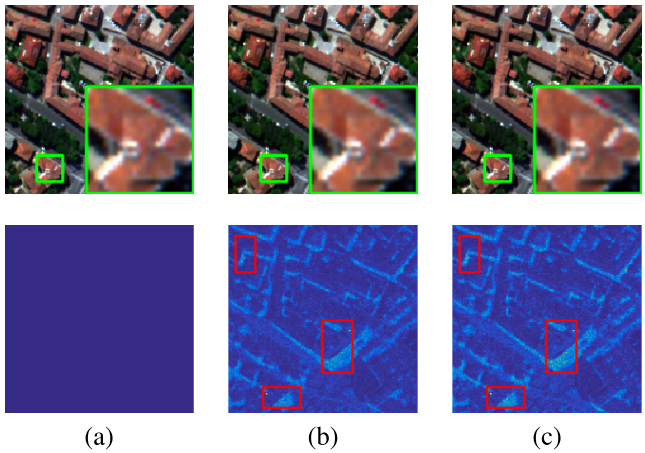


Fig. 11. Example of fusion results for *PC_patch21* (the 21st patch from the PC dataset) illustrating the effectiveness of the lightweight hybrid 3-D-2-D CNN. The top and bottom rows show the reconstructed images with demarcated areas zoomed in 4 times and corresponding visualization of SAM results, respectively. We present the composite HSIs with bands 60-30-10 as R-G-B. (a) Ground truth. (b) Hybrid 3-D-2-D CNN. (c) 2-D-CNN only. Our proposed EC-FTN with the lightweight hybrid 3-D-2-D CNN can obtain better quality of both spatial details reconstruction and spectral information fidelity.

2) Effect of the VGG16 Network: To investigate the validity of employing the VGG16 network for edge map generation, we carry out experiments on the PC dataset without using VGG16 network, i.e., the edge map is obtained by applying the Sobel operator to the RGB version of HR-MSI directly in the experiments. Table IX depicts the average quantitative results of different methods for edge map generation. It can be seen from Table IX that our EC-FTN with VGG16 network can yield superior fusion results compared to the method of directly using the Sobel operator for the generation of $\mathbf{E}_m$. Experimental results in Table IX affirm that the proposed edge map generation method realized by applying the Sobel operator to feature maps resulting from the pretrained VGG16 network is more advantageous for providing the edge prior and facilitating the fusion performance.

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TABLE IX

AVERAGE QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS FOR EDGE MAP GENERATION (WITH AND WITHOUT VGG16 NETWORK) ON THE PC DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
Without VGG16	46.4115	2.5471	0.0063	1.4800	0.9884
With VGG16	**46.5093**	**2.5288**	**0.0062**	**1.4711**	**0.9888**

TABLE X

AVERAGE QUANTITATIVE PERFORMANCE OF DIFFERENT NETWORK DESIGNS (WITH AND WITHOUT EC-FML) ON THE PC DATASET

Method	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
Without EC-FML	46.4025	2.5590	0.0064	1.4894	0.9880
With EC-FML	**46.5093**	**2.5288**	**0.0062**	**1.4711**	**0.9888**

3) Effect of the EC-FML: In order to verify the effectiveness of the proposed EC-FML, we perform experiments on the PC dataset using a variation of our method without any EC-FML in the FFTN, with other settings keeping the same with the proposed EC-FTN. The average quantitative results of different network designs are reported in Table X. It can be seen from Table X that our proposal equipped with EC-FMLs obtains better performance than the variant method in terms of all the quantitative metrics, which proves that the proposed EC-FML incorporating the edge detail information adaptively is effective for further improving the fusion accuracy.

4) Effect of the Weighted Combinatorial Loss Function: To demonstrate the effectiveness of the proposed $\mathcal{L}_{\mathrm{WC}}$ loss, we evaluate the fusion performance of our EC-FTN trained with different loss functions on the PC dataset. Regarding the proposed network using only the standard MAE loss as the baseline, we add the L_{IGD} and L_{SA} terms to the L_{MAE} loss, respectively, to investigate the effect of each component in our loss function $\mathcal{L}_{\mathrm{WC}}$. As reported in Table XI, our EC-FTN can provide promising fusion results even with the simple MAE loss, indicating the advantage of our network design. The EC-FTN with L_{MAE} plus L_{IGD} loss can achieve better spatial measures such as PSNR and UIQI than using only the L_{MAE} loss, which validates that the IGD loss is profitable for recovering the high-frequency structure information such as edges. However, the direct addition of L_{IGD} term degrades the spectral quality, which may be due to the over emphasis on the gradient-level loss. In contrast, our model with the combination of the L_{MAE} and L_{SA} loss functions performs better in retaining the spectral characteristics but achieves lower PSNR and UIQI values than the network with L_{MAE} plus L_{IGD} does, which demonstrates the importance of L_{IGD} once again. In addition, it is found that combining the SA loss contributes to the improvement of global fusion accuracy since it obtains a smaller ERGAS value than the single L_{MAE} does. In comparison with the first three variants with different loss functions, the quantitative results obtained by utilizing the proposed $\mathcal{L}_{\mathrm{WC}}$ loss are the best in terms of all the metrics, as summarized in Table XI. The comparison results show that the newly designed weighted combinatorial loss function is able to achieve a satisfactory balance among pixel-level, gradient-level, and spectral constraints, thus demonstrating the validity of our $\mathcal{L}_{\mathrm{WC}}$ loss function.

TABLE XI

AVERAGE QUANTITATIVE PERFORMANCE OF EC-FTN WITH DIFFERENT LOSS FUNCTIONS ON THE PC DATASET

Loss Function	PSNR	SAM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	0	0	1
L_{MAE}	46.4120	2.5676	0.0064	1.4948	0.9879
$L_{MAE} + L_{IGD}$	46.5074	2.5828	0.0063	1.5065	0.9884
$L_{MAE} + L_{SA}$	46.3928	2.5671	0.0063	1.4893	0.9883
$\mathcal{L}_{WC}$	**46.5093**	**2.5288**	**0.0062**	**1.4711**	**0.9888**

V. CONCLUSION

In this work, we propose a novel EC-FTN for HS-MS fusion, which mainly comprises three phases: feature extraction, feature fusion, and transformation and image reconstruction. First, to extract the joint spectral-spatial features of HS and MS images efficiently, we employ 3-D convolutions followed by reshaping layers in the two FENs. This lightweight hybrid 3-D-2-D CNN is able to not only exploit the spatial-spectral property adequately, but also facilitate the training. Then, the extracted features are fully fused and transformed in the FFTN equipped with FMLs to incorporate the edge detail information adaptively, which is conditioned on the edge map generated via TL. Finally, the HR-HSI is reconstructed from the transformed features through IRN. In addition, a new weighted combinatorial loss function is designed to guide the training process, which helps to balance the pixel-level, gradient-level, and spectral constraints effectively. Comparative experiments with the state-of-the-art traditional and DL-based approaches on both ground-based and remotely sensed HS datasets validate the superior fusion performance of our newly proposed EC-FTN method, especially in terms of edge details reconstruction.

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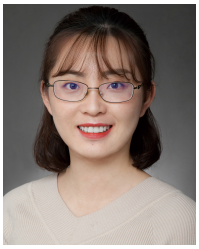
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Yuxuan Zheng (Student Member, IEEE) received the B.E. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where he is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Services Networks.

His research interests include remote sensing data fusion, hyperspectral image super-resolution, and deep learning.



Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an exchange Ph.D. Student of Mississippi State University, Starkville, MS, USA, supervised by Dr. Qian Du. She is an Associate Professor and a Post-Doctoral Researcher with the School of Telecommunications Engineering, Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and computer vision.



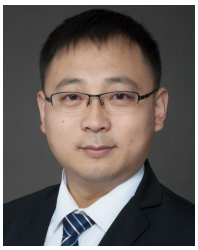
Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He is a Professor with the School of Telecommunications Engineering, Xidian University. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, Xidian University. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.



Jie Guo received the B.S. and Ph.D. degrees in information and communication engineering from Xidian University, Xi'an, China, in 2005 and 2010, respectively.

He is an Associate Researcher with the School of Telecommunications Engineering, Xidian University. His area of research interests include image and video compression, artificial intelligence, deep learning, and the corresponding very large-scale integration (VLSI) implementation.



Xianyun Wu received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2012.

He is an Associate Professror with the State Key Laboratory of Integrated Services Networks, Xidian University. His research interests include remote sensing, image compression and processing, high-performance computing, and distributed source coding.



Yanzi Shi (Student Member, IEEE) received the B.S. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where she is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Services Networks.

Her research interests include hyperspectral target detection, feature extraction, urban remote sensing, and deep learning.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP. He was a Visiting Scholar with Stanford University, Stanford, CA, USA, KTH, Stockholm, Sweden, and National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in Remote Sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. He is a Professor of signal and image processing with Grenoble INP. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008, and the Institut Universitaire de France from 2012 to 2017. He was the Founding President of the IEEE Geoscience and Remote Sensing (GRS) Society French Chapter from 2007 to 2010, which received the 2010 IEEE GRS-S Chapter Excellence Award. He was the Vice-President of the IEEE GRS Society (GRSS), the In-Charge of meetings and symposia from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). He was the Chair from 2009 to 2011 and the Co-Chair of the GRSS Data Fusion Technical Committee from 2005 to 2008. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He is a Highly Cited Researcher (Clarivate Analytics, Thomson Reuters) from 2018 to 2019.

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